

Multiple Choice

1. **Answer:** A

Options: A) **; B) //; C) is; D) not in

Note: In Python 3, the double asterisk operator (**) is specifically designated for performing exponentiation, allowing for the calculation of a number raised to the power of another number.

2. **Answer:** A

Options: A) A group of cookies stored by the user's Web browser; B) The Internet Protocol (IP) address of the user's computer; C) The user's e-mail address; D) The user's public key used for encryption

Note: A group of cookies stored by the user's Web browser can track and store a wide range of personal data, including browsing history and preferences, which can be accessed by third parties, thereby significantly compromising the user's personal privacy.

3. **Answer:** A

Options: A) Round-robin; B) Priority queuing; C) Shortest job first; D) Youngest job first

Note: The round-robin scheduling policy is starvation-free because it allocates a fixed time slice to each job in a cyclic order, ensuring that all jobs receive regular service and preventing any job from waiting indefinitely.

4. **Answer:** C

Options: A) 17_{16} ; B) E_4_{16} ; C) E_7_{16} ; D) F_4_{16}

Note: The correct answer is E_7_{16} because when converting the decimal number 231 to hexadecimal, it breaks down to 14 (E) in the 16's place and 7 in the 1's place, which together form E_7 .

5. **Answer:** A

Options: A) 1; B) 2; C) 3; D) 4

Note: The correct answer is 1 because the `min()` function in Python 3 returns the smallest element from an iterable, and in the list `l = [1, 2, 3, 4]`, the smallest value is 1.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The set of Boolean operators {AND, OR} is not complete because it lacks the

ability to express negation, which is essential for representing all possible Boolean expressions.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the expression $x \gg 1$ with x set to 8 evaluates to 4, not 3, as a right bitwise shift by 1 divides the number by 2.

8. **Answer:** False

Options: A) True; B) False

Note: The maximum possible degree of the minimal-degree interpolating polynomial $p(x)$ for $n + 1$ distinct points is n , as a polynomial of degree n can uniquely interpolate $n + 1$ points only if the points lie on a polynomial of degree at most n .

9. **Answer:** False

Options: A) True; B) False

Note: The language $\{ww \mid w \in (0 + 1)^*\}$ is not context-free because it requires the ability to compare two potentially unbounded segments of a string for equality, which cannot be accomplished using a pushdown automaton that has only a single stack.

10. **Answer:** False

Options: A) True; B) False

Note: Local variables are typically stored on the stack and are not included in the activation record frame, which primarily contains information such as return addresses, parameters, and saved registers.

Long Form Response

11. **Answer:** `def babylonian_squareroot(number): | return 0; | return g;`

Note: The answer correctly identifies the need for a function that computes square roots using the Babylonian method, although it lacks the necessary iterative calculation steps and initialization of the variable 'g' to approximate the square root.

12. **Answer:** `def common_prefix_util(str 1, str 2): | return (result) | def common_prefix(arr, n): | return (prefix)`

Note: The provided answer correctly outlines the structure of a function that utilizes a helper function to compare pairs of strings and extract the longest common prefix from an array of strings, which is essential for solving the problem effectively.

End of Answer Key